Lecture 3: Numerical Descriptive Techniques

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Textbook: Keller, G. & Gaciu, N. (2020). Statistics for Management and Economics, 2nd ed., Cengage, ISBN: 9781473768260.

Recap - The Big Picture

Remember our map of Statistics from last week?

So far, we have been working with **Descriptive Statistics**.

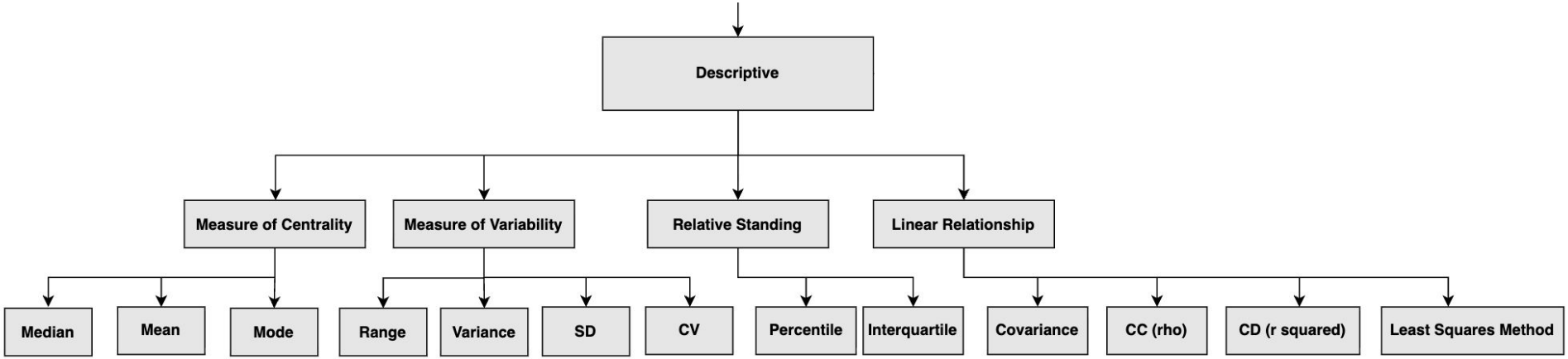
- **Chapter 2:** How to organize data
- **Chapter 3:** How to summarize data visually

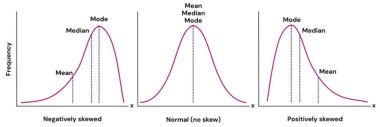
Chapter 4: How to summarize data numerically

- **This chapter is about numerical descriptive statistics.** We use numbers to describe the center, spread, and relative variability of data.

The Goal:

Transform raw data into actionable business information.


Descriptive
Measure of Centrality
Measure of Variability
Relative Standing
Linear Relationship
Median
Mean
Mode
Range
Variance
SD
CV
Percentile
Interquartile
Covariance
CC (rho)
CD (r squared)
Least Squares Method


Frequency
Mode
Median
Mean
x
Negatively skewed
Mean
Median
Mode
x
Normal (no skew)
Mode
Median
Mean
x
Positively skewed

Same units as data
Squared units
Easier to interpret
More mathematically useful
Standard Deviation
Variance

Goals Scored Over the Last 7 Games

1 3 4 6 6 7 8

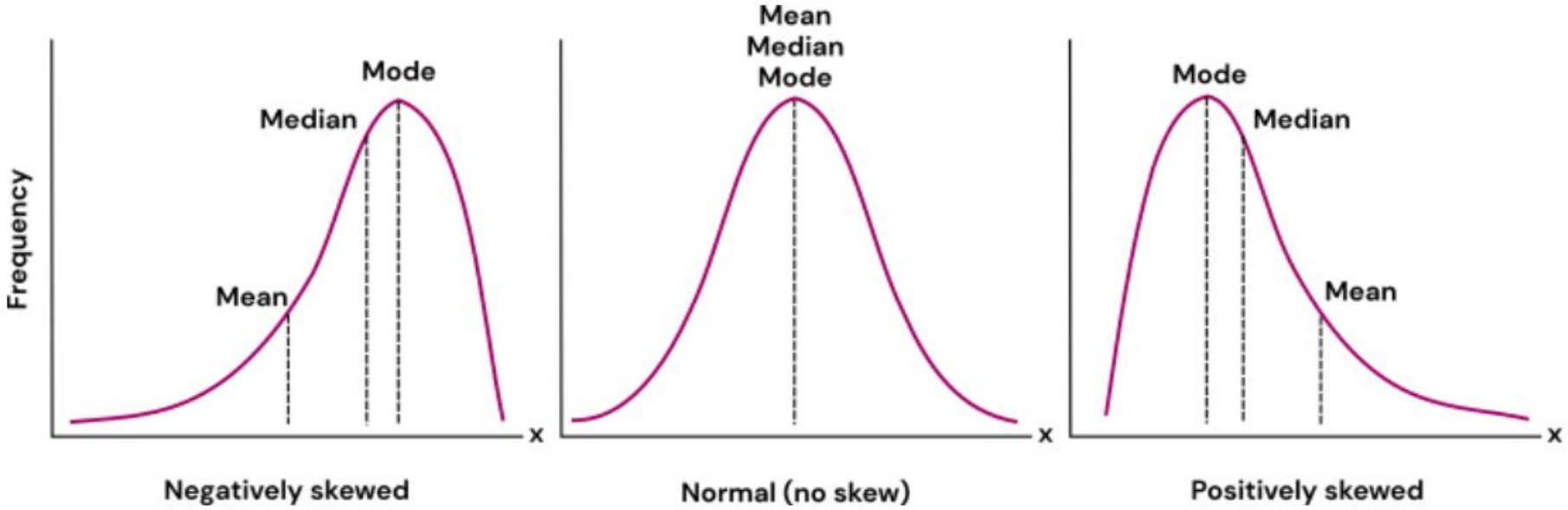
mean 5
average

mode 6
most common

median 6
middle

range 7
largest - smallest

Mean, Median, Mode: Which is Best?



Example: Calculating Variance and Standard Deviation

Suppose we have the following sample data:

$$5,\ 7,\ 9$$

Step 1: Calculate the mean

$$\bar{x} = \frac{5+7+9}{3} = 7$$

Step 2: Find deviations from the mean

$$5-7=-2, \quad 7-7=0, \quad 9-7=2$$

Step 3: Square the deviations

$$(-2)^2 = 4, \quad 0^2 = 0, \quad 2^2 = 4$$

Step 4: Calculate the sample variance

$$s^2 = \frac{4+0+4}{3-1} = \frac{8}{2} = 4$$

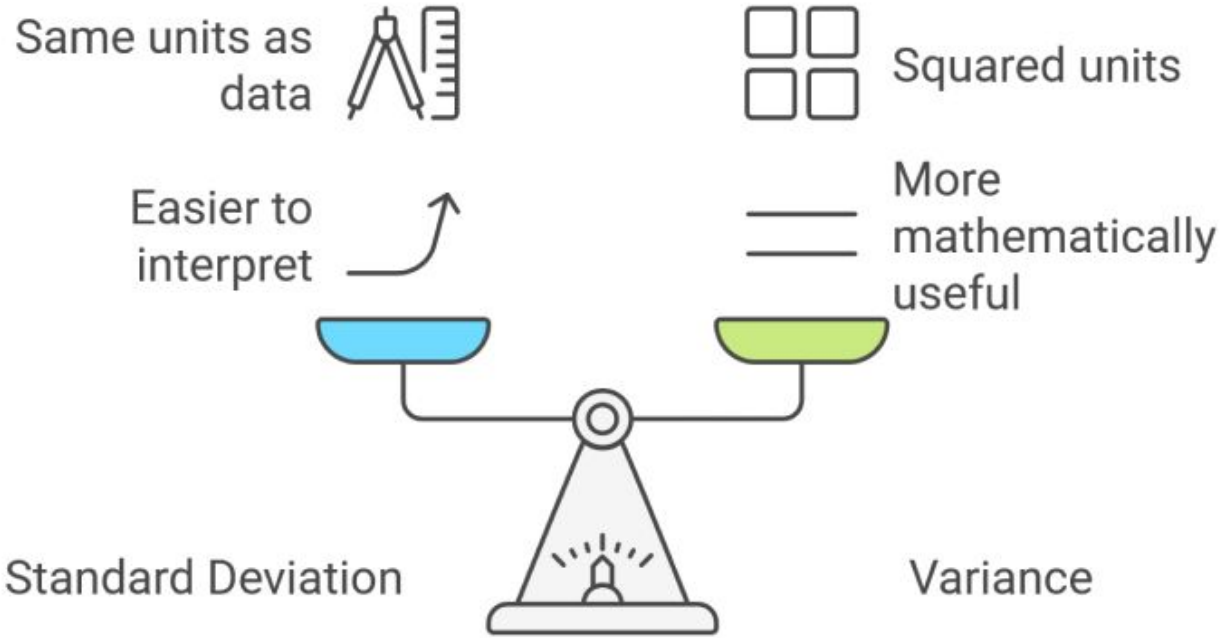
Step 5: Calculate the sample standard deviation

$$s = \sqrt{4} = 2$$

Variance and Standard Deviation

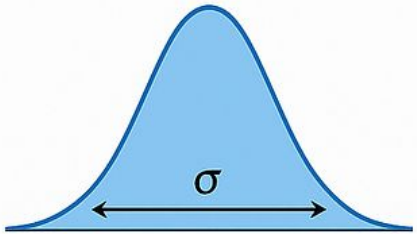
	Population	Sample
Variance	$\sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$	$S^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$
Standard deviation	$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$	$S = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}}$

Variance Vs. Standard Deviation



STANDARD DEVIATION VS STANDARD ERROR VS VARIANCE

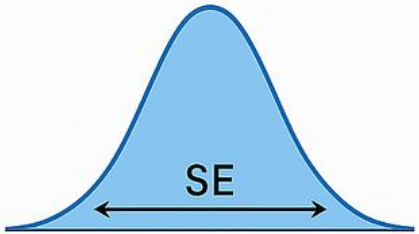
STANDARD DEVIATION



$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

Spread of individual data points

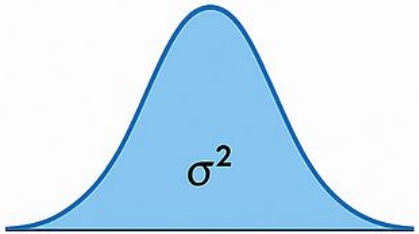
STANDARD ERROR



$$SE = \frac{\sigma}{\sqrt{n}}$$

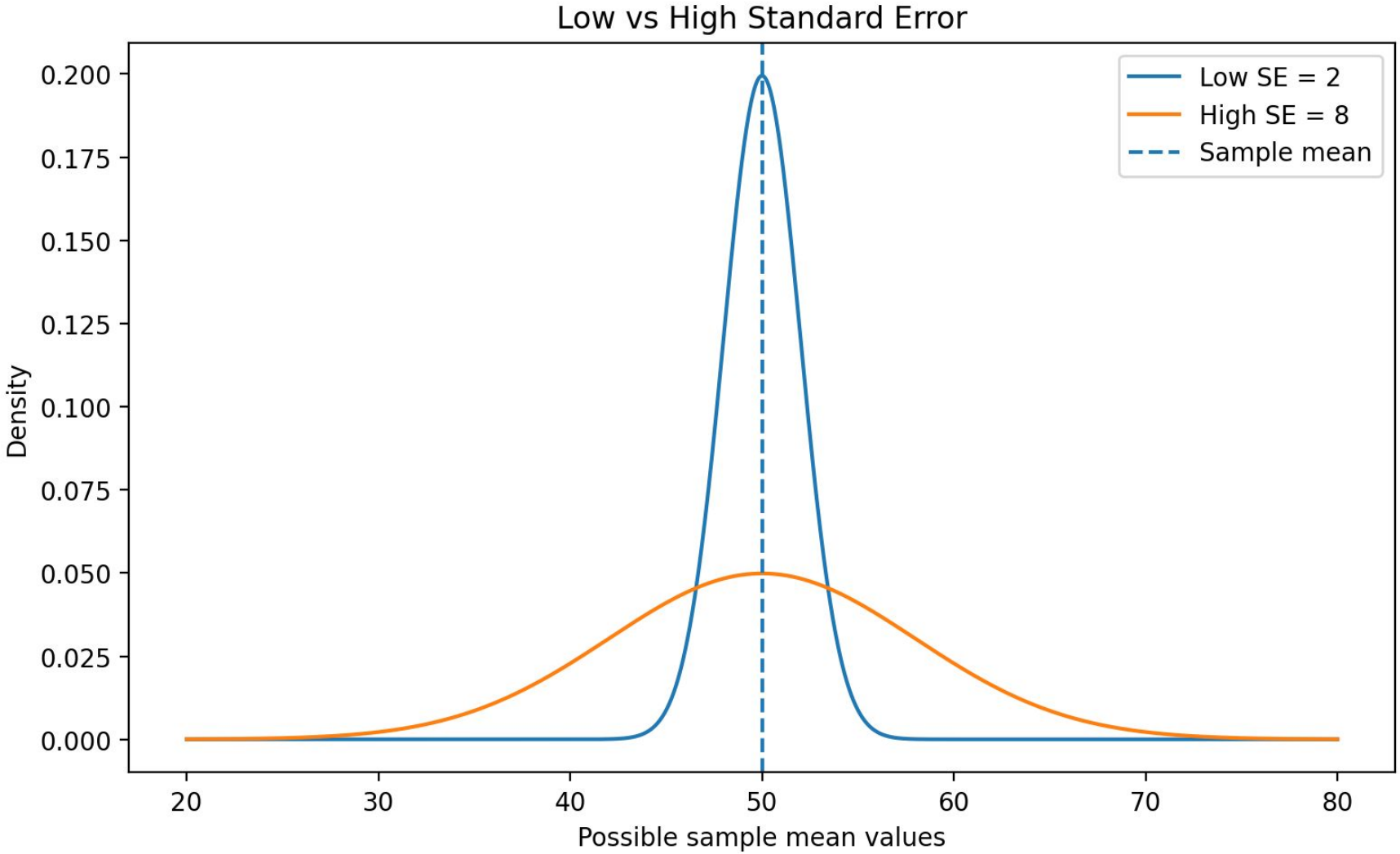
Precision of sample mean estimate

VARIANCE



$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

Average squared deviation


Low vs High Standard Error
Low SE = 2
High SE = 8
Sample mean
Density
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
20
30
40
50
60
70
80
Possible sample mean values

Coefficient of Variance

Coefficient of Variation:

Stock A: CV = 100%

(5/5*100=100%)

Stock B: CV = 5%

(5/100*100=5%)

$$CV = \left(\frac{S}{\overline{X}}\right) \cdot 100\%$$

Measures of Position

n = number of interval

n = 4

QUARTILE

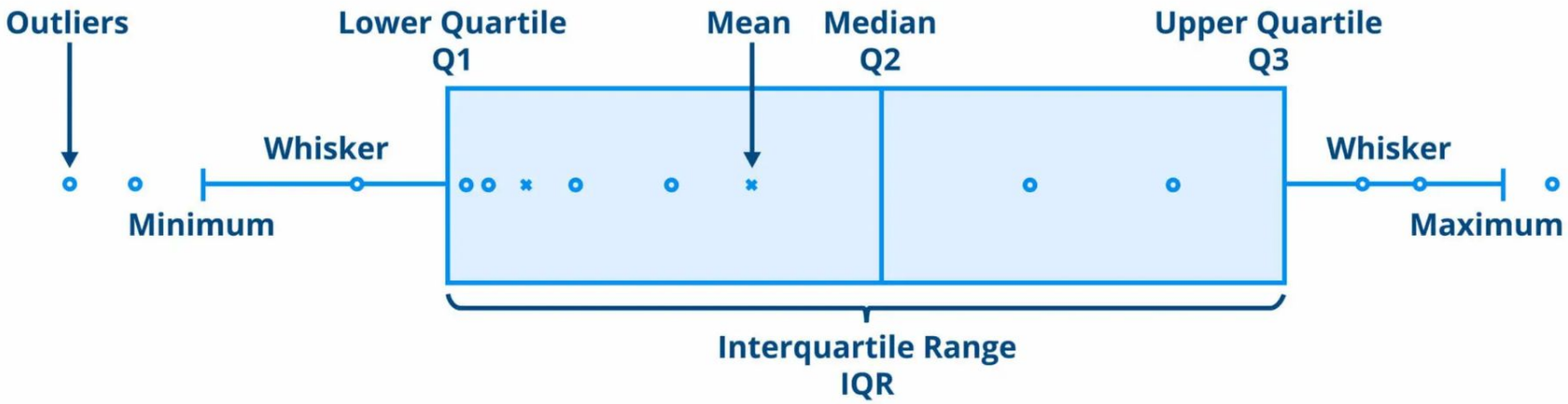
n = 10

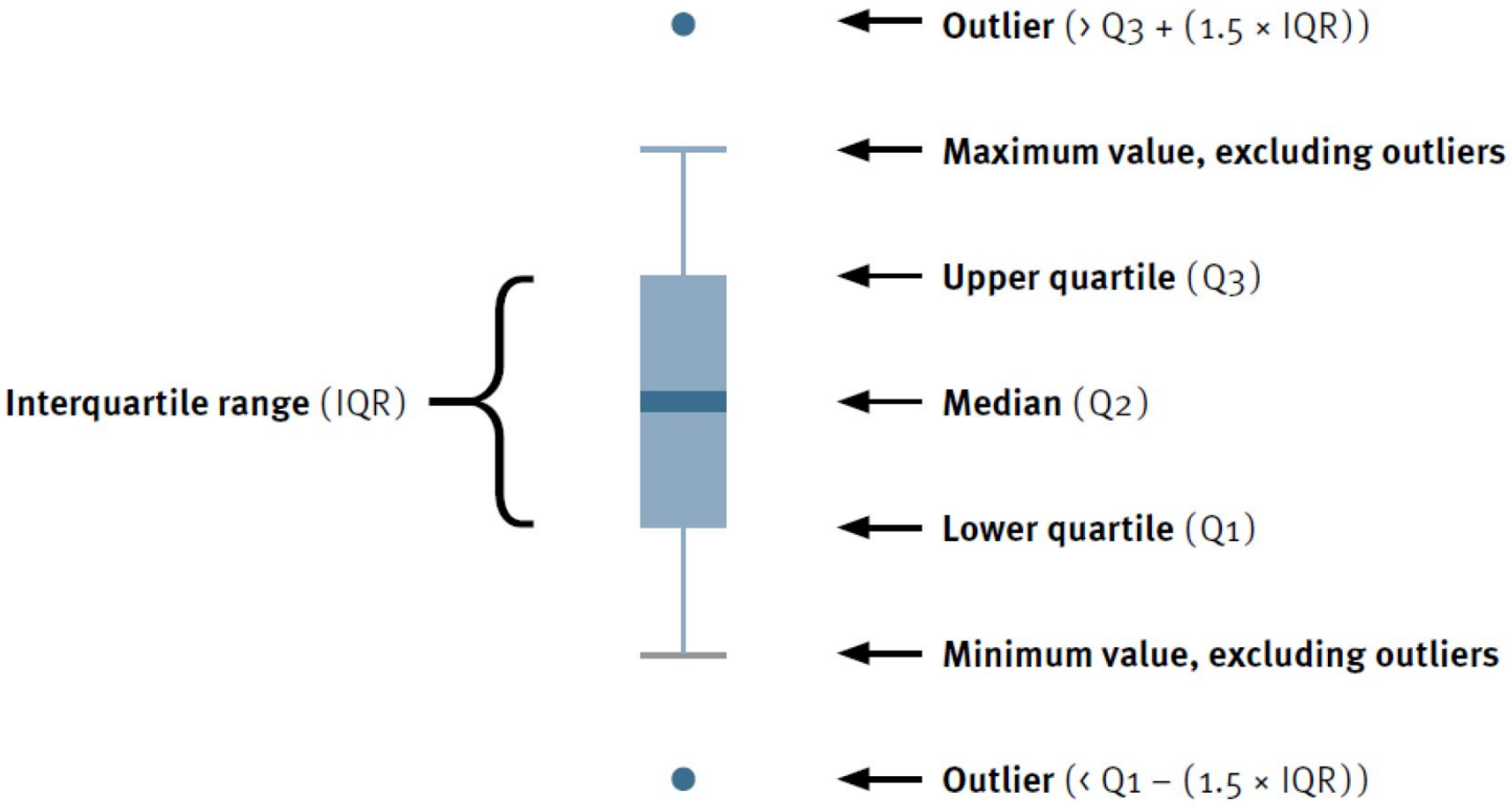
DECILE

n = 100

PERCENTILE

Box plot


Outliers
Lower Quartile
Q1
Mean
Median
Q2
Upper Quartile
Q3
Whisker
Whisker
Minimum
Maximum
Interquartile Range
IQR


Outlier (› Q3 + (1.5 × IQR))
Maximum value, excluding outliers
Upper quartile (Q3)
Interquartile range (IQR)
Median (Q2)
Lower quartile (Q1)
Minimum value, excluding outliers
Outlier (‹ Q1 – (1.5 × IQR))

What is the Degrees of Freedom

Degree of freedom (DF) is a term that refers to the number of independent variables or observations that are available in a statistical model or analysis. In the context of linear regression, degree of freedom refers to the number of observations or data points that are used to fit the model, and it plays a crucial role in determining the precision of the model's predictions. There are two types of degree of freedom in linear regression:

- Residual degree of freedom
- Total degree of freedom

Degrees of Freedom

How many independent pieces of information you really have.

Example: 5 test scores with an average of 70

- Pick any 4 scores you want.
- The 5th score is no longer free — it must make the average stay 70.
- So only 4 scores can vary freely.
- That means the degrees of freedom are 4.

Average fixed at 70

Must be whatever makes the average stay 70

69	71	70	68	?
free	*free*	*free*	*free*	*forced*

Choose these freely

5 values − 1 fixed average = 4 degrees of freedom

Degrees of freedom = number of values − number of constraints

Degrees of Freedom

How many independent pieces of information you really have.

Example: 5 test scores with an average of 70

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Average fixed at 70

Must be whatever makes the average stay 70

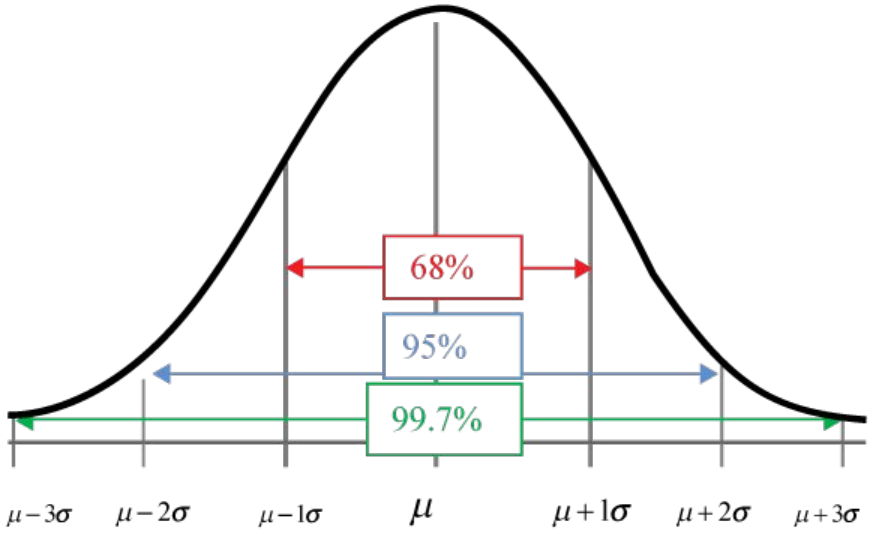
69	71	70	68	72
free	*free*	*free*	*free*	*forced*

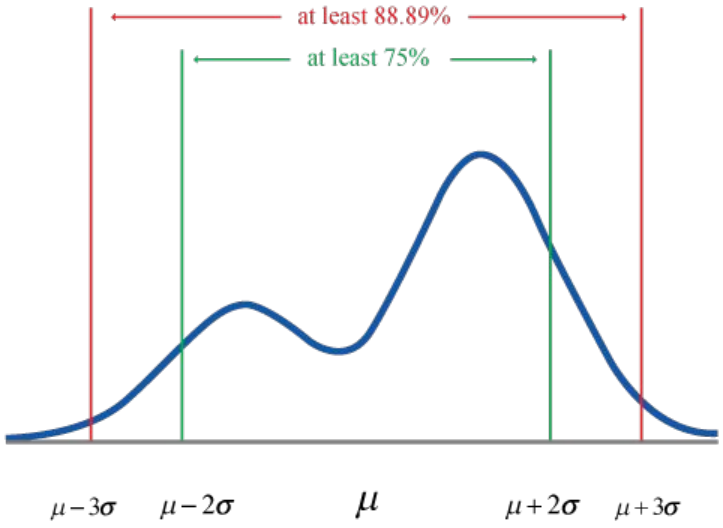
Choose these freely

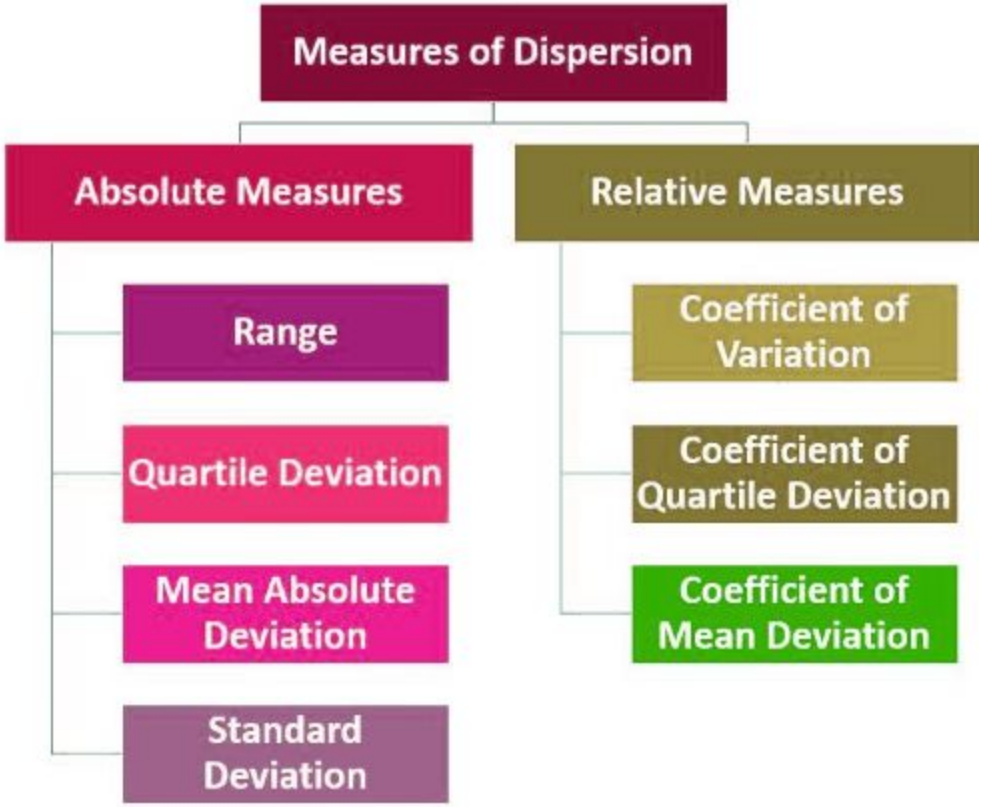
5 values − 1 fixed average = 4 degrees of freedom

Degrees of freedom = number of values − number of constraints

Chebyshev's Theorem






Measures of Dispersion
Absolute Measures
Relative Measures
Range
Quartile Deviation
Mean Absolute Deviation
Standard Deviation
Coefficient of Variation
Coefficient of Quartile Deviation
Coefficient of Mean Deviation